MACHINE LEARNING BASELINES

COVID-19 Chest X-Ray Survival Prediction

Pretrained DenseNet features, evaluated across geographical regions

Problem statement

Can chest X-ray features predict patient survival, and does that hold in a region the model has never seen?

Primary task: binary survival classification from X-ray-derived features, evaluated with leave-one-region-out validation.

Dataset

Input

Chest X-rays, anteroposterior (AP) or posteroanterior (PA) views only

Metadata

Clinical findings, survival status, patient information, image view, region

Target

Binary survival label: survived or did not survive

Source

Public COVID-19 Chest X-ray Dataset, loaded through TorchXRayVision

Eligibility filtering

1. Survival status is known
2. No intubation present
3. Image offset greater than 0 and less than 8 days

Only records meeting every criterion are used, so the final sample size depends on missing values. Records with an unknown region are excluded from the region-aware classifier benchmark.

Feature-extraction pipeline

1. Preprocess

Center crop and resize to 224 × 224

2. Extract

Pretrained DenseNet from TorchXRayVision, inference only

3. Collect

Intermediate features and pathology predictions

4. Classify

Conventional classifiers on the extracted features

No CNN is trained from scratch; the pretrained representations are reused for the downstream task.

Representations compared

Intermediate DenseNet features

Image vectors taken from inside the network

Pathology outputs

The model's predictions for chest radiographic pathologies

Selected pathology outputs

A chosen subset of those predictions

Class-prevalence baseline

Uses only target-class frequency, as the reference point

The goal is to see whether image-derived features add information beyond a simple baseline.

Validation and classifiers

Leave-one-region-out

Each fold holds out one region for testing and trains on the others. This tests cross-region generalization, not a random split of examples.

Four classifiers

- Logistic regression
- RBF support vector classifier
- Extra Trees
- Random Forest

Linear and SVM models use standardized features; tree models use 500 estimators with class weighting. Metrics: AUROC and AUPRC.

Results: region-aware benchmark

Classifier	Mean AUROC	Mean AUPRC
Extra Trees	0.697	0.943
Random Forest	0.683	0.935
Logistic Regression	0.678	0.936
RBF Support Vector Classifier	0.564	0.895

Reading the numbers

0.697

Extra Trees mean AUROC

0.943

Extra Trees mean AUPRC

AUPRC can stay high when the positive class is common, so the moderate AUROC is the better signal: survival discrimination across regions remains challenging.

Limitations and future work

Limitations

- Few held-out regions and eligible patients
- Performance may vary by region
- Does not show X-rays alone give reliable survival prediction
- Exploratory, not clinical readiness

Next steps

- Larger, more representative datasets
- Patient-level deduplication
- Wider validation and uncertainty estimation
- Bias assessment and independent external validation